

Institute of Actuaries of India

Subject CA1-II – Actuarial Risk Management

October 2015 Examinations

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

Policy data

Checks for accuracy:

- Physical verification - Regular checks
 - a. Carry out a few sample checks of information as contained in actual proposal forms, claims forms with information as captured in policy administration system. This should be carried out regularly rather than once a year
- Check whether each policy record satisfy boundary conditions for eg.-
 - a. Minimum/Maximum age at entry
 - b. Minimum/Maximum policy term,
 - c. Minimum/Maximum premium payment term,
 - d. Maximum maturity age,
 - e. Minimum/Maximum SA
 - f. Minimum/Maximum Premium amount
- Check for unusual values for eg., 0 SA, 0 premium, or premium greater than maximum allowed
- Ratios not consistent; premium per 1000 SA
- Check for blanks- for eg. Blank DOB, blank SA, blank premium, blank term etc
- Check out for wrongly entered values- for eg. Gender not captured as M or F, incorrect DOB, Date of birth > date of policy commencement, date of valuation
- Consistency checks :
 - a. Average premium, average sum assured, average premium per 1000 sa of current investigation should be consistent and sensible with the figures of previous investigation.
- Checks for Completeness:

Reconciliation

 - a. Prepare movement data over the inter-valuation period. Reconcile the number of policies, Sum assured, Premium inforce at the end of the year using the previous data and movements data
 - b. Reconcile certain payments with accounting data- for eg., benefit paid with accounting data
 - c. Reconcile premiums received on new policies and renewal policies with accounting data

[7]

Solution 2:

i) The assumptions required for valuation would be:

- Interest Rate
- Mortality assumptions
- Morbidity assumptions (if any accelerated CI benefit or riders covered)
- Future maintenance Expenses, claims and investment expenses
- Expense inflation
- Withdrawal assumptions (if to use in the reserve calculations)

[3]

ii)

While determining each of the assumptions, certain basic principles need to be considered:

- It should have regard to any guidelines by the regulator or actuarial profession
- Assumptions should be prudent

Mortality/Morbidity:

- As it is a new company, there is no past experience
- Also since the product is recently priced, can use pricing assumption with margins for prudence
- Normally for term products, mortality would be varied by gender, smoker/non-smoker, rural/urban etc. If this has been done at pricing the same should be used in reserving as well

Interest rate:

- As it is a term assurance product, interest rate will not be a major assumption, could still be significant assumption for single pay term and limited pay term policies
- This rate could be chosen with reference to yields on the assets supporting the liabilities.

Withdrawal:

- Withdrawal assumption should be taken in the valuation only if it increases the reserve. If by having the withdrawal assumption reserves would decrease, then caution needs to be used while setting the withdrawal assumption for reserving
- Can consider Withdrawal assumption assumed in the pricing the product
- When setting withdrawal assumption allowance may also need to be made for selective withdrawals as early withdrawals are likely from healthy policyholders which would mean that the mortality of the remaining policyholders could be worse than expected.

Expenses

- Since it is a new company actual maintenance expenses etc may be higher than break even expense assumption made at the time of pricing.

- Local regulations or actuarial guidance may require actuaries to make provision for expected future maintenance overruns as well
- Projections may be needed for staff, systems, overheads etc. and future new business to arrive at the expected per policy expenses.
- Alternatively, assumptions used in pricing stage can also be used with adjustments for prudence.
- More margins in expenses may be required for prudence as it can very sensitive to the future volume of business (which is unknown and it is a relatively new company)

Expense Inflation

- This will depend on expected future earnings inflation. This will depend on the expected future salary increases and other related costs and its impact on policy servicing and claims expenses.
- It must be consistent with the investment return assumption.

[7]

[10 Marks]

Solution 3:

i)

- It is normal for the reinsurance premiums to be more than the reinsurance claims over a longer period, as we expect the reinsurers to load for their profits and expenses.
- ... the reinsurer has to make money to be in this business
- That the claims have been lower could be because there have been no industrial disasters, epidemics, geographical disasters over the last 3 years and this is just a matter of chance
- Reinsurance premiums are based on confidence that the reinsurer has with the direct insurers underwriting ability and risk selection and
- company's ability to do business and achieve business targets (more business the cedant writes higher volumes for reinsurer as well on the quota share business)
- reinsurance markets are going through a hard cycle and hence premiums are generally high
- the information available with the CFO about market experience may not be correct or could be incomplete
- other market participants may have a different reinsurance agreement which costs less, for eg., no XL cover
- other market participants may have opted for reinsurers with lower credit rating than the company which could also lead to a lower costs even if programs are similar
- Group business tends to create aggregation of risks as all employees are in one location; reinsurance provides cover against aggregation of claims or catastrophe claims, hence it is unfair to compare premiums and claims received over such a short period.

- There could be other payables received from reinsurers for eg the reinsurance commissions. Hence the comparison could be misleading.

[5]

ii)

To decide on whether the existing treaties can be continued requires modeling.

- Project the future cashflows over the period of reinsurance or a period of five to ten years.
- As the projections will be sensitive to the claims, a stochastic model needs to be adopted with mortality being modeled stochastically.
- For a definite reinsurance program which is a combination of Quota share and XL and the chosen retention levels and chosen projection period, project the company's balance sheet and profit and loss account for the period decided including all reinsurance inflows and outflows.
- As the business being projected is group term assurances which are generally one year renewable, the expected business year on year needs to be arrived.
- Projections need to be realistic so the assumptions used must be on a best estimate basis.
- Criteria for testing should be arrived at and it depends on the risk appetite and other factors such as free reserves of the company.
- Criteria for testing could be probability of ruin or any other criteria such as loss in any period (for eg. Year) does not exceed a proportion of earnings of business.
- Run atleast 1000 simulations and note down the output of solvency position of the company.
- Check whether for this quota share proportion/retention limit, the criteria are met such as company is solvent or earnings stay above a certain level, for 995 (say) out of 1000 runs.
- Analyse the output parameters like mean, variance etc.
- Sensitivity analysis may be required for varying other parameters in the model
- If the criteria are not met then it may require that the treaties have to be revisited.

[7]

[12 Marks]

Solution 4:

- i) Requirements of a 'Good Model' are:
- Model being used must be valid
 - rigorous enough for the purpose and
 - adequately documented
 - It should be capable of adequately reflecting the risk profile of the product being modeled
 - Parameters used must allow for all the features of the business being modeled,

- It should be appropriate to the business being modeled
- ... take into account any special features of the provider.
- Model should be easy to explain.
- The model should exhibit sensible joint behavior of model variables
- The outputs from the model should be capable of independent verifications for reasonableness
- Model should not be overly complex
- ... making the results difficult to interpret and communicate
- Model should not be too long and expensive to run
- It should be capable of development and refinement

[5]

ii) A model should try to capture

- The characteristics of the different types of claims
- ... short tailed ones like property damage and
- Long tailed ones like liability claims
- Number of claims
- Claim amounts
- How number and size varies with time
- ... due to inflation or
- ... any seasonal effects (more accidents in rainy season)
- Inflation will vary by type of claim
- Accumulation of claims
- Very large individual claims
- Catastrophes
- Reinsurance programs (gross and net claims amounts)

[5]

iii) Interactions include

- Interest Rates and inflation (effecting the claim amounts)
- Economic conditions and claim frequency
 - a. Recession may be associated with more claims
 - b. ... increased crime or fraudulent claims
 - c. ... customers may claim even for small amounts
- Economic conditions and claim severity
 - a. Again recession may lead to the claim severity increased

- b. Claim amounts would be inflated
- Geographic location and claims
 - a. Natural calamities
 - b. Weather related claims (rainy seasons, floods related damage, severe winter)

[5]

[15 Marks]

Solution 5:

i)

There may not be a single reason for the general insurance company to go below the minimum solvency requirement. It could be a combination of some of the following reasons including mismanagement and non availability of proper risk management process in the company

- Excessive adverse claims due to actual experience being very different from the assumed such as :
 - a. too much accumulation of risks especially- properties in same geographical location
 - b. poor underwriting standards leading to anti-selection and hence higher than expected claims.
 - c. policy wording not clear and exclusions or benefit limits not properly written.
- Reserving not sufficient to meet the liabilities.
- Inadequate and inappropriate reinsurance.
- Failure of any third party like brokers, external vendors
- Catastrophic losses much higher than expected and the CAT cover proving inadequate (CAT covers are not unlimited)
- Natural disasters like earthquake, flooding, tsunami etc. which would get compounded if portfolio is not geographically diversified
- Drastic fall in business – This could also be due to aggressive competitors or competitors bringing in new innovative products and the company missing out on the opportunity.
- Under- pricing of products either intentionally or unintentionally and excessively selling such products.
- Reinsurers default
- Fraud by persons within the organisation, outsiders and policyholders
- Mismanagement
- Adverse changes due to Economic recession leading to high number of claims and increased severity, frauds
- Investment strategy failure and excessive losses for eg. Aggressive investment strategy without sufficient mitigation process and/or without sufficient free reserves
- Parent or subsidiary company's failure
- Bad name/drastic fall in credit rating for company due to some recent developments leading to loss of business

- Sudden changes in regulatory requirement for eg. Increasing reserving requirement [9]

ii) Steps regulator can put in place:

Controls and monitoring

- Put in place risk based supervision and that companies have robust Risk management process and usage of proper risk management tools
- Have regular disclosures and reports to the regulator- for eg., financial statements, solvency report etc.
- Regulate assets that they can invest in
- ... have less risky assets so that risk of insolvency can be reduced
- ... have exposure limits on counterparties and various investment assets
- Requirement for companies to provide a detailed report on claims management process- for eg. Case reserving policy, rules on setting up of IBNR
- Mandate investigations at regularly frequency on
 - a. Claims
 - b. Investment
 - c. Asset liability management
 - d. Expenses
- Minimum credit rating/capital requirement for Reinsurance companies that a company can place business with
- Approve pricing of products to ensure they are appropriate priced
- Also ask company to submit and monitor the sensitivity of the capital to various stress situations at least annually
- Board approved risk management policy, underwriting policy, asset liability matching policy, reinsurance policy etc. to be submitted to regulators yearly.
- To ensure diversification of risk e.g. Limits on maximum percentages of business in one geographical location, single property etc.

Governance

- As part of good governance encourage or mandate companies to
- Have continuous audit of all their processes and financials
- Peer review of actuarial activities like pricing, modeling, reserving, etc
- ensure each company has whistle blower policy in place.

Penalties

- Have severe penalties for breaching any of the above controls or limits
- Additional solvency requirements incase
- Asset liability mismatch
- Non-diversification of business; any aggregation of risks

Regulator not to wait till the solvency capital reaches minimum levels. The process should start when the company is nearing even say 150% of the minimum solvency levels.

So that it is not too late to take any corrective steps.

Company should be asked to investigate the reasons
... recovery plan to be presented
... closer reporting of implementation of the plan and its success.

[9]

Solution 6:

i) Risks in the project:

- Large initial expenditure, the inflow will come later on.
- It will take many years to build; and then operated to generate revenue
- Construction company should be able to manage this cashflow; no net positive cashflows till many years
- Asset created is physical, hence it is
 - a. Indivisible
 - b. Unique
 - c. unmarketable
- The road not getting built on time
- Technical glitches in making the road; like a tunnel may need to be built to create a way
- Any sub-contractors not meeting their commitments
- More maintenance and repair costs due to poor quality of the expressway
- Any financial irregularities
- Not able to collect enough toll over 25 years
 - ... route not popular with vehicles
 - ... alternate route better, cheaper
 - ... railways between those cities may be faster, cheaper and more convenient
 - ... New means of transport making this route unpopular. For eg. Metro rails. 25 years is a too long a period any change can happen.
 - ...one of the cities becoming less popular
 - ...Due to climatic changes, the route or the cities becoming more prone to natural disasters.
- Too many accidents on the road, making it unpopular with users
- Government changes and revoke the contract; so not recovered full costs yet
- Government agencies may want to have a say on the amount of toll charged; which may make the project financially unviable
- Workers agitation (any issues relating to working conditions, wages, safety, etc)

[9]

ii) Insurance covers the company can take are:

Employer's Liability – indemnifies the employer in case of any accidental injury or death of an employee due to negligence of the employer or other employee during the construction and maintenance

Public Liability – indemnifies the insured against any legal liability for the bodily injury or death or damage to property of a third party (other than employees) due to negligence of employer for e.g., damage to nearby property or body of 3rd party during construction and maintenance

Professional indemnity – There will be many technical professionals involved in the whole project including engineers. Indemnifies the company against legal liability resulting from negligence in the provision of a service by professionals of the company.

Property contents – indemnifies against value of loss or damage of contents/equipments like due to theft, accident, rains, floods, fire, etc. which are crucial for construction. Also any major equipments malfunctioning or major repairs should be covered.

Property buildings – again indemnifies against value of loss or damage of office or homes of laborers due to theft, accident, rains, floods, fire, etc.

Pecuniary loss – protects against any bad debts, third party failures, sub-contractors failure

Fidelity guarantee – protects from any financial loss due to fraud, embezzlement by employees

Business interruption – as it is a project going on for a reasonably long time, don't want any interruptions due to fire, agitations, strikes, etc

Personal accident – fixed benefit on any disability due to accident of employees

Health insurance (Medical insurance) – provides cover for any medical treatment for its employees

Group Term Insurance – to provide cover for all its employees and even contract workers

Key man Insurance – for its main employees

[10]

[19 Marks]

Solution 7:

i) Advantages of disclosure to employees:

- Disclosure leads to transparency and employees can scrutinize and feel more confident about the security of their savings.

- This is a DC plan. Hence employees may like to know how their funds are performing on a regular basis
- Also it is a contributory scheme, employees can make decisions on when and how much to contribute to their pension fund based on fund performance
- Employees are made aware of the any potential problems for eg., fund performance is volatile or has taken risky investment positions. They can take up with the corresponding persons for corrective actions.
- If the employees have an option to choose or change the investment strategy, this disclosure will help them to make a decision.
- Employees can at any point in time assess their benefit entitlements and plan for their financial needs.
- Employees can understand the terms offered by the insurance company and plan for retirement accordingly
- Employees are reminded of the paternalistic view of the company from time to time and it helps to retain employees.
- If employees are unhappy with the scheme, they can leave the scheme.

[4]

ii) General information:

- Contributions:
 - a. contributions percentage/ amount made by employees and employer during the period (in the current year/ till date).
 - b. if contribution depends on salary then the definition of salary used for contribution purposes.
- Benefit entitlements:
 - a. Eligibility criteria
 - b. Benefits at the time of exit
 - c. in case of withdrawal, retirement etc.
 - d. Annuity options available, lump sum benefits payable, etc
 - e. whether employees can transfer accumulated amount on resignation etc.
- Investment strategy employed
 - a. If the employees have an option to choose the investment type the options available and the risks involved with each option.
 - b. The past performance and charges in each of the fund options
- Expense charges (if any).
 - a. For administration
 - b. For fund management

- c. it may be % of fund or fixed amount.
- Treatment of entitlements in the event of insolvency. For eg. Whether employees can transfer the fund to another scheme or whether the accumulations will continue in scheme without further contributions etc.

Member-wise details

- Contributions by Employees
- Contributions by employer
- Accumulated amount at the time of disclosure.
- Likely accumulated amount at the time of retirement based on assumptions of salary, investment return. Variability of accumulations by changing the assumptions can be given.
- Likely annuity amounts for standard annuity options can be given. Again variability of annuity can also be given by varying the annuity rate assumptions.

[8]

iii) Factors to consider for the trustees to make the decision

- It is true that the insurance company will load for its profits and if there is no competition the profits loaded could be high.
- But the insurance company might be charging less due to economies of scale.
- There is also a risk that if the insurance company fails, the responsibility of paying the annuity may come back to the scheme.

Factors for self managing the annuities

- It should be seen whether it is permissible according to the countries legislation,
- There could be a requirement that the annuity should be necessarily purchased from the insurance company. If so, then the director's suggestion cannot be accepted.
- They may internally have or need to hire actuaries to assist in the annuity rate calculations
 - ... which need to be closely monitored for changing investment and other factors
 - ...this could be expensive as an actuarial consultant needs to be engaged
- For very large schemes:
 - ... the scheme is already managing the accumulation part of the scheme, so managing the annuity part will not be a huge additional administrative cost/ effort
 - ...there may be economies of scale
 - ... will have investment expertise
- It may be an additional expense for smaller or medium sized funds

- Unless for very large schemes, the fund may not achieve the pooling of risk
- ... towards the longevity risk
- ... in most cases it may be better to transfer that risk to an insurer, as they are experts and have a larger portfolio to diversify their risk.

[7]

[19 Marks]
